

Safety and Outcomes of Dabigatran Reversal With Idarucizumab Before IV Thrombolysis in Patients With Acute Ischemic Stroke

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Abstract

Background and Objectives

IV thrombolysis (IVT) is contraindicated in patients with acute ischemic stroke (AIS) while on dabigatran or other oral anticoagulant (OAC) treatment. Idarucizumab completely reverses the effect of dabigatran within minutes, without increasing the risk of thromboembolism. Limited data exist on IVT treatment after dabigatran reversal with idarucizumab. We aimed to investigate the safety and outcomes of IVT after dabigatran reversal in patients with AIS.

Methods

This is an observational study based on the Safe Implementation of Treatment in Stroke (SITS) International Stroke Thrombolysis Registry. Hospitals treating patients with AIS contributed data. The main outcome was safety as measured by any parenchymal hematoma (PH), symptomatic intracerebral hemorrhage (SICH) per SITS, and death within 3 months. The secondary outcome was functional independence defined as modified Rankin Scale scores of 0–2 at 3 months. Propensity score matching (PSM) was used to compare patients treated with dabigatran reversal to patients without prior OAC in primary analysis and for secondary analysis between patients treated with dabigatran reversal with idarucizumab to patients on dabigatran without reversal treatment.

Results

Among 258,589 IVT treated patients with AIS, 510 were on dabigatran and 156 received dabigatran reversal. Dabigatran reversal patients were older (75 vs 69 years, $p < 0.01$), had similar median baseline NIH Stroke Scale (9 vs 10, $p = 0.47$), and longer onset to IVT time (185 vs 150 minutes, $p < 0.01$) compared with patients without prior OAC treatment ($n = 191,648$). After PSM analysis where good balance was achieved, patients treated with dabigatran reversal before IVT had similar results in all outcomes as compared with patients treated with IVT without prior OAC (any PH: 3 vs 9%, $p = 0.10$; SICH: 1 vs 1%, $p = 1.00$; death: 25 vs 19%, $p = 0.33$, functional independence: 51 vs 52%, $p = 0.95$). Secondary analysis showed similar results for all outcomes before and after PSM.

Discussion

In our observational study with patients with AIS, IVT treatment after dabigatran reversal was safe and had similar outcomes to IVT treatment without previous OAC. Furthermore, comparison between dabigatran reversal vs no reversal also indicates that IVT is safe for patients on dabigatran without idarucizumab reversal, which is currently being investigated in clinical trials.

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Class of Evidence

Criteria for rating therapeutic and diagnostic studies

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Supplementary Material

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Glossary

AIS = acute ischemic stroke; **DOAC** = direct OAC; **DRI** = dabigatran reversal with idarucizumab; **ECASS** = European Cooperative Acute Stroke Study; **EVT** = endovascular thrombectomy; **IVT** = IV thrombolysis; **mRS** = modified Rankin Scale; **NIHSS** = NIH Stroke Scale; **OAC** = oral anticoagulant; **PH** = parenchymal hematoma; **PSM** = propensity score matching; **SICH** = symptomatic intracerebral hemorrhage; **SITS** = Safe Implementation of Treatment in Stroke; **SITS-ISTR** = SITS International Stroke Thrombolysis Registry; **SITS-MOST** = SITS Monitoring Study.

Classification of Evidence

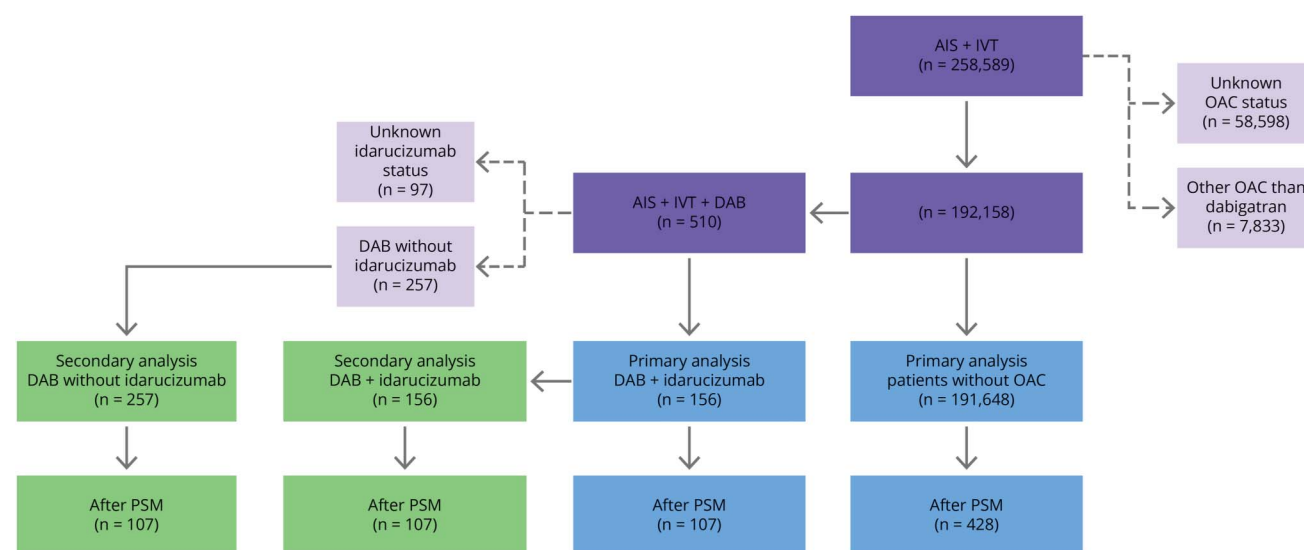
This study provides Class III evidence that IVT after idarucizumab reversal in patients with AIS who took dabigatran within 48 hours of onset of symptoms is safe and comparable with those of patients with AIS not on prior oral anticoagulation.

Introduction

Dabigatran is the first approved non-vitamin K oral anticoagulant directly inhibiting thrombin.¹ According to international guidelines, IV thrombolysis (IVT) is contraindicated in patients suffering an acute ischemic stroke (AIS) with an intake of dabigatran within 48 hours.^{2,3} However, the American Stroke Association states that IVT may be considered if renal function and coagulation laboratory tests are normal.³ The European Stroke Organization also refers to the uncertainty of testing thrombin time in regard to the effect of dabigatran and furthermore to the reversal of dabigatran with idarucizumab before IVT treatment as questionable due to sparse research at present.² However, the Australian and New Zealand stroke guidelines recommend dabigatran reversal with idarucizumab (DRI) before IVT, based on expert consensus.⁴

Idarucizumab is a humanized monoclonal antibody fragment which completely reverses the effect of dabigatran within minutes, without increasing the risk of thromboembolism.⁵ Prior mostly single nation observational studies, systematic reviews, and case series have indicated that IVT in patients with AIS may be safe and effective after DRI.⁶⁻¹² However, the total number of patients in these studies has been limited, and more real-world data would increase clinician confidence regarding the safety of dabigatran reversal before IVT in patients with AIS. Our primary aim was to investigate the safety and outcomes of dabigatran reversal in patients with AIS treated with IVT within an international registry. Our secondary aim was to investigate safety and outcome of IVT in patients with AIS taking dabigatran who received IVT after dabigatran reversal compared with no reversal agent. We aimed to investigate the safety and outcomes of IVT after DRI compared with patients with AIS not on prior oral anticoagulation.

Figure Study Flowchart



AIS = acute ischemic stroke; DAB = dabigatran; IVT = IV thrombolysis; OAC = oral anticoagulant; PSM = propensity score matching.

Table 1 Baseline and Demographic Characteristics Before PSM for the Primary Study Population

	DRI (n = 156)	No OAC (n = 191,648)	p Value
Previous atrial fibrillation	86.2 (100/116)	18 (30,017/166,877)	0
Age	75.8 (9.7)	69.9 (13.7)	0
Systolic blood pressure (mm Hg), baseline	150.7 (27.9)	152.9 (24.2)	0.34
Diastolic blood pressure (mm Hg), baseline	84.8 (15.2)	83.5 (14.5)	0.277
Glucose (mmol/L), baseline	8.6 (8.1)	7.7 (5.9)	0.203
Onset to door (min)	95.5 (60–135)	79 (54–120)	0.006
Onset to imaging (min)	119 (83–169)	106 (75–149)	0.078
Onset to IVT (min)	185.5 (135–225)	150 (114–192)	0
Door to needle (min)	75 (48–109.5)	60 (38–90)	0
NIHSS score, baseline	9 (5–16)	10 (6–16)	0.471
mRS score, baseline	0 (0–1)	0 (0–1)	0
Sex, male	55.1 (86/156)	54.6 (104,633/191,645)	0.958
Previous hypertension	81.9 (127/155)	67.3 (126,844/188,425)	0
Previous hyperlipidemia	45.4 (69/152)	30.7 (56,147/182,963)	0
Previous diabetes mellitus	23.1 (36/156)	20.2 (38,425/190,563)	0.42
Previous congestive heart failure	15.8 (24/152)	7.6 (14,268/187,361)	0
Previous TIA	11 (17/154)	5.8 (8,763/152,008)	0.008
Previous Stroke	32.1 (50/156)	11.9 (22,635/190,251)	0
Smoking, current and previous	23.1 (33/143)	29.1 (52,168/179,442)	0.137
Antiplatelet treatment, baseline	11.5 (18/156)	37.4 (71,509/191,173)	0
Antihypertensive treatment, baseline	76.1 (108/142)	58.9 (105,753/179,605)	0
Statin treatment, baseline	39 (60/154)	27.5 (42,381/153,907)	0.002
Antidiabetic treatment, baseline	17.6 (25/142)	16.4 (23,535/143,305)	0.79
EVT treatment	23.1 (36/156)	11.8 (22,538/191,648)	0

Abbreviations: EVT = endovascular thrombectomy; IVT = IV thrombolysis; mRS = modified Rankin Scale; n = number of patients; NIHSS = NIH Stroke Scale; OAC = oral anticoagulant; PSM = propensity score matching. Mean (SD), median (interquartile range), and percentage (number of cases/number of patients with data) presented for continuous, ordinal, and categorical variables, respectively. *p* Values calculated by Student *t* test, Mann-Whitney *U* test, and Pearson χ^2 test for continuous, ordinal, and categorical variable, respectively.

Methods

Study Design

This is an observational registry study where data were collected from Safe Implementation of Treatment in Stroke International Stroke Thrombolysis Registry (SITS-ISTR), which is a global stroke treatment registry. The registry has multiple data entry forms with different entry titles such as, IVT, endovascular thrombectomy (EVT)–treated patients. The registry also has variables regarding details on atrial fibrillation, oral anticoagulant (OAC) treatment, type of OAC such as direct OAC (DOAC), or dabigatran reversal. Furthermore, baseline and demographic characteristics were collected. Follow-up at

24 hours, 7 days/discharge, and a final follow at 3 months were performed, as previously reported.^{13,14} Imaging studies were performed at baseline and 22–36 hours after IVT treatment where radiologic assessment was performed locally according to clinical routine at centers participating in the SITS-ISTR.

Study Population

We included patients with AIS treated with IVT, with either on-going dabigatran therapy or no OAC treatment before stroke onset. We then identified patients with AIS on dabigatran treatment before stroke onset who received idarucizumab for reversal therapy. Our primary study population compared IVT-treated patients with AIS with DRI to patients

Table 2 Baseline and Demographic Characteristics After PSM for Primary Study Population

	DRI (n = 107)	No OAC (n = 428)	p Value
Atrial fibrillation	86 (92/107)	86 (368/428)	1
Age	76 (9.1)	77.1 (10.1)	0.29
Systolic blood pressure (mm Hg)	151 (29.6)	155.2 (23.9)	0.18
Diastolic blood pressure (mm Hg)	84.6 (15.4)	84.2 (15.3)	0.82
Glucose (mmol/L)	7.4 (2.6)	7.5 (2.5)	0.72
Onset to door (min)	96.5 (55.5–133.25)	85 (57.75–132.25)	0.86
Onset to imaging (min)	116 (73–164.5)	117 (85–163)	0.29
Onset to IVT (min)	180 (128.5–220)	164.5 (120–210.25)	0.37
Door to needle (min)	73 (45–103.5)	60 (40–93)	0.03
NIHSS score	10 (5–16)	9.5 (5–15)	0.77
mRS score	0 (0–1)	0 (0–1)	0.37
Sex, male	52.3 (56/107)	43.5 (186/428)	0.12
Hypertension	83.2 (89/107)	74.9 (319/426)	0.09
Hyperlipidemia	35.2 (37/105)	30.8 (127/413)	0.44
Diabetes mellitus	17.8 (19/107)	20.4 (87/427)	0.64
Congestive heart failure	17.8 (19/107)	16 (68/425)	0.77
Previous TIA	11.2 (12/107)	5.5 (19/343)	0.07
Previous stroke	24.3 (26/107)	13.9 (59/424)	0.01
Smoking, current and previous	21.6 (21/97)	19.7 (80/407)	0.76
Antiplatelet treatment	10.3 (11/107)	49.8 (213/428)	<0.01
Antihypertensive treatment	77.5 (79/102)	72.2 (288/399)	0.34
Statin treatment	31.1 (33/106)	25.4 (87/342)	0.30
Antidiabetic treatment	11.8 (12/102)	14.6 (46/314)	0.57
EVT treatment	24.3 (26/107)	20.3 (87/428)	0.44

Abbreviations: DRI = dabigatran reversal with idarucizumab; EVT = endovascular thrombectomy; IVT = IV thrombolysis; mRS = modified Rankin scale; n = number of patients; NIHSS = NIH Stroke Scale; OAC = oral anticoagulant; PSM = propensity score matching. Mean (SD), median (interquartile range), and percentage (number of cases/number of patients with data) presented for continuous, ordinal, and categorical variables, respectively. *p* Values calculated by Student *t* test, Mann-Whitney *U* test, and Pearson χ^2 test for continuous, ordinal, and categorical variable, respectively.

with AIS without any recent intake of OAC before stroke onset. Our secondary study population included only patients with recent intake of dabigatran treatment, where we compared dabigatran reversal vs no reversal treatment. Patients with other OAC treatment before stroke onset and those with incomplete data were excluded.

Outcomes

Main outcomes were safety as measured by any parenchymal hematoma (PH), symptomatic intracerebral hemorrhage (SICH) per SITS Monitoring Study (SITS-MOST) (local or remote parenchymal hemorrhage type 2 at 22–36 hour follow-up CT scan, and neurologic deterioration of ≥ 4 NIH Stroke Scale (NIHSS) points from baseline

or the lowest score during the first 24 hours after treatment, or death) definition, and death within 3 months. The secondary outcome was functional independence defined as modified Rankin Scale (mRS) scores of 0–2 at 3 months.

Statistical Analysis

Descriptive statistics were performed for baseline and demographic characteristics to compare between groups. For outcome analyses, we performed propensity score matching (PSM) to adjust for baseline imbalances and to reduce eventual biases. We included key variables in the PSM based on potential confounding effects, leading to the following list of variables: age, baseline NIHSS score, baseline mRS score,

Table 3 Outcomes After PSM for Primary Study Population

	DRI (n = 107)	No OAC (n = 428)	p Value
SICH by SITS-MOST	1 (1/103)	0.7 (3/412)	1
SICH by ECASS2	2.9 (3/103)	2.9 (12/414)	1
mRS 0–2 at 3-mo follow-up	50.9 (29/57)	52.3 (157/300)	0.95
Death by 3-mo follow-up	25.4 (15/59)	18.9 (58/307)	0.33
Any PH	3 (3/99)	8.6 (34/395)	0.1
PH1	1 (1/99)	2.8 (11/395)	0.51
PH2	1 (1/99)	2.8 (11/395)	0.51
PHr1	NA (0/99)	2.5 (10/394)	0.23
PHr2	2 (2/99)	1 (4/394)	0.76

Abbreviations: DRI = dabigatran reversal with idarucizumab; ECASS = European Cooperative Acute Stroke Study; n = number of patients; OAC = oral anticoagulant; PH = parenchymal hemorrhage; PSM = propensity score matching; SICH = symptomatic intracranial hemorrhage; SITS-MOST = Safe Implementation of Treatment in Stroke Monitoring Study. Percentage (number of cases/number of patients with data) presented. p Values per Pearson's χ^2 test.

atrial fibrillation, onset-to-IVT time, and treatment with EVT. The corresponding propensity score was calculated for each subject and a nearest neighbor matching algorithm was used within $0.2 \times \text{SD}$ of the logit of the propensity score. To determine whether the propensity score approach achieved balance in the selected variables, we compared the same baseline and demographic characteristics after PSM as we did before PSM. For the primary analysis, we used a 1:4 ratio allocation of cases. For the secondary analysis, we used a 1:1 ratio allocation of cases.

A p-value of <0.05 was considered statistically significant. All statistical analyses were performed in the software R version 4.3.0 (The R Foundation for Statistical Computing Platform, Vienna, Austria).

Standard Protocol Approvals, Registrations, and Patient Consents

Requirements for ethical approval and patient consent for participation in the SITS-ISTR differed among participating countries. Ethical approval and patient consent were obtained in countries that required this, while other countries approved the register for conduct as an anonymized audit. The SITS Register was approved by the Research Ethics Committee in Stockholm, Sweden, any subsequent amendment by the Swedish National Ethics Review Authority (Dnr 2015/767-31, later amendment 2022-01157-02).

Data Availability

Access to the anonymized SITS-ISTR data will be available from the corresponding author upon reasonable request from qualified researchers, contingent on approval by the SITS Scientific Committee.

Results

We identified 258,589 patients in the SITS registry with AIS who were treated with IVT (Figure). Of these, we identified 510 patients who were on dabigatran treatment before stroke onset, of which 156 received DRI and 97 were excluded due to unknown DRI status. We also identified 191,648 IVT treated patients with AIS with no recent intake of any OAC before stroke. In addition, 7,833 were excluded due to OAC treatment other than dabigatran before stroke onset, and 58,598 were excluded due to unknown status of recent OAC intake.

Primary Analysis

Table 1 presents baseline and demographic characteristics before PSM for the population for the primary analysis, which included 156 patients with dabigatran reversal and 191,648 patients without any prior OAC treatment. Patients treated with dabigatran reversal compared with no OAC treatment before stroke onset were older (76 vs 70 years), had similar baseline NIHSS scores (9 vs 10), longer onset to IVT treatment times (186 vs 150 minutes), higher proportion of atrial fibrillation (86% vs 18%), lower proportion of antiplatelet treatment (12% vs 37%), and higher proportions of cardiovascular risk factors including previous stroke (32% vs 12%), hypertension (80% vs 67%), and hyperlipidemia (45% vs 31%). eTable 1 presents the outcomes before PSM, and there were no statistically significant differences between groups for any of the results, with only a trend for higher rates of 3-month mortality (24% vs 16%, $p = 0.05$).

Table 2 presents the baseline and demographic characteristics after PSM, which included 107 dabigatran reversal patients and 428 patients with no OAC use before stroke onset. Both groups were well-balanced for the matched variables after PSM. After PSM, dabigatran reversal patients had longer door to needle times (73 vs 60 minutes), higher proportion of previous stroke (24 vs 14%), and lower proportion of antiplatelet treatment (10 vs 50%).

Table 3 presents outcomes after PSM; there were no statistically significant differences in outcomes between the 2 groups, with only a trend for a lower proportion of any PH for dabigatran reversal patients as compared with no OAC patients (3% vs 9%) (Table 2).

Secondary Analysis

For the secondary analysis, we included 156 patients with dabigatran reversal and 257 patients with no reversal with idarucizumab (eTable 2). Dabigatran reversal patients as compared with patients without reversal with idarucizumab had longer onset to IVT treatment time (186 vs 165 minutes) and door to needle time (75 vs 62 minutes). There was no difference in outcomes between groups before PSM (eTable 3).

After PSM, we included 107 patients from both groups. Dabigatran reversal patients, as compared with patients

Table 4 Baseline and Demographic Characteristics After PSM for the Secondary Study Population

	Idarucizumab (n = 107)	No idarucizumab (n = 107)	p Value
Atrial fibrillation	86 (92/107)	86 (92/107)	0.97
Age	76 (9.1)	76.1 (10.5)	1
Systolic blood pressure (mm Hg)	151 (29.6)	151.6 (23.8)	0.87
Diastolic blood pressure (mm Hg)	84.6 (15.4)	84.6 (13.2)	0.99
Glucose (mmol/L)	7.4 (2.6)	7.7 (2.6)	0.39
Onset to door (min)	96.5 (55.5–133.25)	90 (63.5–137.5)	0.4
Onset to imaging (min)	116 (73–164.5)	110 (82–166)	0.61
Onset to IVT (min)	180 (128.5–220)	165 (104.5–219)	0.31
Door to needle (min)	73 (45–103.5)	56 (29.5–91)	0.02
NIHSS score	10 (5–16)	11 (5.5–17)	0.71
mRS score	0 (0–1)	0 (0–1)	0.49
Sex, male	52.3 (56/107)	57.9 (62/107)	0.49
Hypertension	83.2 (89/107)	87.9 (94/107)	0.44
Hyperlipidemia	35.2 (37/105)	44.2 (46/104)	0.24
Diabetes mellitus	17.8 (19/107)	31.8 (34/107)	0.03
Congestive heart failure	17.8 (19/107)	20.8 (22/106)	0.70
Previous TIA	11.2 (12/107)	9.4 (10/106)	0.84
Previous stroke	24.3 (26/107)	24.3 (26/107)	1
Smoking, current and previous	21.6 (21/97)	14.7 (15/102)	0.28
Antiplatelet treatment	10.3 (11/107)	8.4 (9/107)	0.81
Antihypertensive treatment	77.5 (79/102)	84.2 (85/101)	0.30
Statin treatment	31.1 (33/106)	33.3 (35/105)	0.85
Antidiabetic treatment	11.8 (12/102)	26.7 (27/101)	0.01
EVT treatment	24.3 (26/107)	24.3 (26/107)	1

Abbreviations: EVT = endovascular thrombectomy; IVT = IV thrombolysis; mRS = modified Rankin scale; n = number of patients; NIHSS = NIH Stroke Scale; OAC = oral anticoagulant; PSM = propensity score matching. Mean (SD), median (interquartile range), and percentage (number of cases/number of patients with data) presented for continuous, ordinal, and categorical variables, respectively. *p* Values calculated by Student *t* test, Mann-Whitney *U* test, and Pearson χ^2 test for continuous, ordinal, and categorical variable, respectively.

without reversal with idarucizumab, still showed longer door to needle time (73 vs 56 minutes) and a lower proportion of diabetes mellitus (18% vs 32%) (Table 4). Furthermore, there were no differences between the 2 groups for any of the outcomes after PSM (Table 5).

Classification of Evidence

This study provides Class III evidence that IVT after idarucizumab reversal in patients with AIS who took dabigatran within 48 hours of onset of symptoms is safe and comparable to those of patients with AIS not on prior oral anticoagulation.

Discussion

Our study showed no differences in safety and outcomes of IVT treatment between patients with AIS with recent intake of dabigatran who received reversal with idarucizumab compared with those without any ongoing OAC pretreatment. Furthermore, we found no difference in safety and outcomes when comparing IVT treatment for patients with AIS with recent intake of dabigatran who received reversal with idarucizumab vs no reversal with idarucizumab. Our combined results suggest that IVT is safe in patients with AIS taking dabigatran before stroke regardless of reversal with idarucizumab.

Table 5 Outcomes After PSM for Secondary Study Population

	Idarucizumab (n = 107)	No idarucizumab (n = 107)	p Value
SICH by SITS-MOST	1 (1/103)	2 (2/100)	0.98
SICH by ECASS2	2.9 (3/103)	3 (3/100)	1
mRS 0–2 at 3-mo follow-up	50.9 (29/57)	55.2 (37/67)	0.76
Death by 3-mo follow-up	25.4 (15/59)	23.6 (17/72)	0.971
Any PH	3 (3/99)	5.3 (5/95)	0.67
PH1	1 (1/99)	1.1 (1/95)	1
PH2	1 (1/99)	1.1 (1/95)	1
PHr1	0 (0/99)	3.2 (3/95)	0.23
PHr2	2 (2/99)	1.1 (1/95)	1

Abbreviations: ECASS = European Cooperative Acute Stroke Study; n = number of patients; OAC = oral anticoagulant; PH = parenchymal hemorrhage; PSM = propensity score matching; SICH = symptomatic intracranial hemorrhage; SITS-MOST = Safe Implementation of Treatment in Stroke Monitoring Study. Percentage (number of cases/number of patients with data) presented. *p* Values per Pearson χ^2 test.

At present, international guidelines do not recommend IVT to patients with AIS on DOAC taken within 48 hours, with weak recommendations for dabigatran reversal before IVT treatment.^{2,3} Previous observational studies have shown results from mostly small study populations from a national level as opposed to international cohorts, limiting the generalizability of these studies.^{9,11,15,16} To date, there is no RCT evaluating IVT in patients with AIS receiving dabigatran reversal. Our study is consistent with earlier research regarding safety of IVT in patients with AIS receiving reversal with idarucizumab,⁵ particularly SICH after dabigatran reversal and IVT. A systematic review from 2021 from Frol et al.⁹ and a national registry study and meta-analysis from Theodorou et al.¹¹ from 2024 reported similar findings, with relatively low rates of SICH in IVT treatment following dabigatran reversal. Our results for SICH by SITS-MOST definition at 1.3% were lower likely due to more liberal SICH definition in previous studies, with our SICH by European Cooperative Acute Stroke Study (ECASS) II definition more closely resembling results in recent studies using similar SICH definitions.¹⁷ Furthermore, previous research has suggested that treatment with IVT in patients with ongoing DOAC therapy appears to be safe without reversal of DOAC treatment, something that our study reinforces in our results from the secondary analysis.^{6,10,12,17} Our secondary analysis of patients on dabigatran with dabigatran reversal vs no idarucizumab showed no significant difference for the 2 groups regarding SICH, death and 3-month mRS scores before and after PSM. This further indicates the safety of IVT in patients with AIS with ongoing dabigatran treatment regardless of reversal of dabigatran. A systematic review and meta-analysis from Shahjouei et al. showed that treatment with IVT was safe within 48 hours of DOAC intake, although it differs from our study since dabigatran reversal patients were actively excluded due to earlier reports on hemorrhagic transformation and death.^{7,18,19} Our results suggest no additional safety issues for IVT treatment of

dabigatran reversal patients as compared with those without reversal with idarucizumab. Our results are also in align with a recent cohort study by Meinel et al.²⁰ where they have a similar SICH rate for IVT in patients on DOAC treatment within 48 hours compared with our study, using the ECASS definition.

According to previous clinical studies, there seems to be no effect of idarucizumab on the endogenous thrombin production as well as no procoagulant effect from the reversal agent.^{5,21} The question remains regarding a possible effect of idarucizumab on the coagulation cascade as previous in vitro studies have shown that idarucizumab added to plasma increases thrombin generation, which needs further investigation for an understanding of the clinical relevance.²²

Our study has several limitations. Our data are built on a registry with the possibility of different biases based on the human factor when registering patients. Missing data led to exclusion of patients due to unknown OAC status, which may have had an impact on the result. Furthermore, the low number of patients on dabigatran may lead to a skewed population selection. The results should be interpreted with caution, given that there is a risk of lack of statistical power to detect a significant difference. There are also inherent limitations to the PSM technique itself, such as the possibility of residual confounding leading to over interpreting the results. In addition, hemorrhagic outcomes were a rare event in all analyzed subgroups, which could mean that our study lacked power to find any statistically significant difference. The main strength of our study is a large international dataset. Our results were consistent regardless of whether IVT was given after dabigatran reversal or without any reversal which further suggests safety of IVT in patients with AIS with recent intake of dabigatran. The strengths of the study are that it provides data regarding IVT treatment to patients on dabigatran with

and without receiving idarucizumab, in addition to the generalizability due to the international study cohort. Combined, these data could help guide physicians while awaiting future RCTs.

Our findings suggest that patients with AIS on dabigatran treatment receiving IVT after reversal with idarucizumab had comparable safety and outcomes as patients with no recent intake of OAC. Similarly, among dabigatran treated patients suffering an AIS, outcomes did not differ after IVT treatment between those with reversal and those without reversal with idarucizumab. While awaiting randomized controlled trial data, our observational findings may assist clinicians in applying an individualized risk-benefit approach to IVT in patients with AIS treated with dabigatran.

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Author Contributions

M. Sälund: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. M. Matusевич: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. G. Tsigoulis: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. A. Chiti: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. S. Frol: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. A. Kiamili: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. A. Peeters: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. M. Petruzzellis: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. E. Rota: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. D. Toni: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. T.P. Moreira: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. N. Ahmed: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data.

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